



Prediction on High-Dimensional Clinical Data

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Abstract— *Predictive modeling in the domain of Healthcare is growing area of AI which enables the healthcare professionals in informed decision-making and greatly helps with treating various diseases. This work focuses on prediction using high-dimensional low sample space (HDLSS) healthcare and biomedical data comprising of patient data dealing with clinical, laboratory, as well as the patient demographic information, psychological measures, questionnaire responses to predict Tinnitus Handicap Inventory (THI) scores. The study explores how leveraging Machine Learning, feature selectors, tabular pretrained models, and advanced ensemble methods improve the prediction compared to traditional Machine Learning approaches and directly affecting the quality of treatments. This work also focuses on tailored and thorough pre-processing which has also directly impacted the quality of our results. Based on our findings, these advanced techniques, and ensemble approaches has surpassed even the State-of-the-Art models like XGBoost.*

Keywords— Clinical data, Tabular data, High-dimensional data, Machine learning, DeepFS, GRACES, TabPFN, Prediction, Healthcare analytics

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1 Introduction

Digital systems are core components of every hospital and research center today. These systems create and store automated data in large quantities. The data are organized in formats that are processed by a computer, corresponding to an extensive data table. Such data originate from various routine health care operations, including, but not limited to, electronic medical records, lab test report, and patient questionnaires about their symptoms and experiences. Therefore, health care systems have automated data collection from various sources, with the data being uniformly structured in tables to enhance analysis and sharing.

The number of high-dimensional, low sample size (HDLSS) scenarios is growing in the biomedical field, especially in studies involving electronic health records (EHRs) and high-throughput omics datasets.^{1,2} Such studies have many more measured variables than samples, resulting in sparse complex data that can overwhelm standard computational methods, and firm power/sample size estimation becomes practically impossible.¹ With data being this unbalanced, the most typical consequence is model overfitting, in which predictive models capture random noise or mere technical artifacts rather than the underlying biological phenomena (if any), and produce results that are highly variable with the introduction of any slight change in the data, which solves the open reproducibility problem at the study.² The use of specialized statistical methods becomes imperative.

There have been recent improvements in machine learning, but very few have been able to show consistent prediction accuracy in high-dimensional tabular datasets from the healthcare domain. On the contrary, large language models (LLMs) and other types of models which are predominantly used on unstructured data, tend to have poor predictive power when employed on tabular data. It is, therefore, crucial to develop meaningful approaches for features from high-dimensional tabular data. Effective representation learning is key to improving the interpretability, and predictive accuracy and stability of models. By obtaining compact and discriminative representations of the features, the predictive power of tasks at the classification and regression levels is enhanced, which is critical for assessing clinical risks and biomedical outcomes.

The goal of this project is to develop an integrated pipeline which is fully capable of handling HDLSS datasets in a very unified manner. It is capable of addressing both classification and regression tasks within this framework. To improve model's performance on imbalanced datasets, an oversampling strategy has been implemented that will optionally compensate for unequal class distributions, ensuring that minority classes or outlier data points (for regression use case) are adequately represented during training phase. By combining this specialized preprocessing, feature extraction, and model optimization steps, the pipeline aims to provide a stable, interpretable, and accurate predictions across the 3 high-dimensional biomedical datasets which have been studied in this project.

2 Related Work

A lot of machine learning and deep learning methods have already been applied to tabular datasets, and this project reviewed several of the stronger options for tabular prediction.

TabNet uses a multi-step attention mechanism to decide which features matter most at each decision step. It works directly on raw tabular inputs and uses decision blocks that make its behavior somewhat comparable to how decision trees focus on a subset of features at a time. Unlike trees, it is fully differentiable, which helps with both training and interpretability at the

local and global level. TabNet also includes a self-supervised pretraining approach that learns from unlabeled data, which can improve performance when labeled data is limited.

SAINT (Self-attention and Intersample Attention Transformer) adapts transformers for tabular learning by applying attention across both columns and rows. Alongside feature-level attention, it adds an inter-sample attention component that models relationships between different samples, which can be useful when examples share structure. SAINT also uses contrastive self-supervised pretraining, which helps it learn more stable representations, especially when datasets are small. Reported results show SAINT can outperform standard deep learning baselines and, on average, can compete well with gradient-boosting approaches on tabular benchmarks.

VIME (Value Imputation and Mask Estimation) is built for tabular data with limited labels and missingness. It uses two self-supervised pretext tasks: imputing masked feature values and predicting which entries were masked. The representations learned from unlabeled data are then used to regularize a supervised predictor trained on a small labeled subset. This setup is practical when labels are scarce and the table is sparse or incomplete.

TabPTM (Tabular Pre-Training via Meta-representation) focuses on heterogeneity across tabular datasets. It maps instances into a shared meta-representation space by describing each instance through similarities to nearby neighbors and their labels. The intent is to transfer knowledge across different datasets without requiring fine-tuning on each new dataset. TabPTM has been reported to work for both classification and regression, with the shared space helping generalization across tasks.

TP-BERTa adapts pre-trained language models to tabular prediction. It tokenizes numeric values using Relative Magnitude Tokenization (RMT) so that numbers become tokens that preserve relative scale, and it uses an Intra-Feature Attention (IFA) module to combine feature names with feature values. In reported experiments, TP-BERTa performs competitively with gradient boosting, particularly on tables with many categorical variables, and it can generalize across several biomedical tasks without heavy dataset-specific tuning.

Taken together, these methods shaped the direction of this project because they rely on attention mechanisms, representation learning, and pretraining—three ingredients that matter when the data is high-dimensional, heterogeneous, and limited in sample size. Reviewing their trade-offs also helps justify the design choices in the proposed framework.

2.1 Research gaps

Even though TabNet, SAINT, VIME, TP-BERTa, and TabPTM can perform well on many tabular benchmarks, high-dimensional, low-sample size (HDLSS) biomedical prediction still creates problems that are hard to ignore if the goal is clinical relevance.

Overfitting and weak generalization. In biomedical tables, it is common to have far more features than patients. With only hundreds or a few thousand samples, deep models like TabNet and SAINT can fit idiosyncrasies of a dataset instead of learning patterns that hold up in an independent cohort. That instability makes the predictions harder to trust outside the training dataset and reduces clinical usefulness.

Dependence on feature semantics. TP-BERTa gets part of its strength from using feature names as inputs. That advantage drops when feature names are anonymized or encoded, which happens often in privacy-sensitive healthcare datasets. If the column identifiers do not carry meaning, the model has less signal to transfer across datasets.

Rare and correlated features. Many tabular methods optimize overall accuracy and can underweight low-frequency features, even when those features represent clinically important

events such as rare variants. Masking-based objectives like VIME can also lean heavily on correlations: the model may learn that one feature predicts another, without learning whether either feature has an independent association with the outcome. That can hurt interpretability and makes it harder to justify decisions in a clinical setting.

Heterogeneity and instability across datasets. Biomedical data often comes from multiple sources with different measurement protocols, coding practices, and missingness patterns. Methods like TabPTM try to address this by building a unified meta-representation, but that approach can be sensitive to the size and composition of the labeled subset and to hyperparameter choices. Small changes in the input can lead to large shifts in the learned representation, which hurts reproducibility. On top of that, the lack of publicly available code for TabPTM limits how easily other researchers can validate or extend it.

Compute and implementation cost. Transformer-style tabular models can be expensive to train and tune, especially with very high-dimensional inputs. Memory requirements and training time can rise quickly, which makes these models harder to use in settings where results need to be produced quickly or infrastructure is constrained.

These gaps point to what a practical approach needs: methods that stay reliable with small samples, do not collapse when column names are masked, pay attention to rare but meaningful signals, and remain stable enough to reproduce across cohorts—all without requiring excessive compute.

2.2 Desired model capabilities

This section outlines what the framework should be able to do for HDLSS biomedical prediction. It does not introduce new algorithms on its own; it describes capabilities the framework should enforce through existing modeling and experimental design choices.

Attention to rare signals. The framework should preserve and surface infrequent features that carry clinical meaning, including uncommon mutations or low-prevalence indicators that may strongly affect outcomes.

Handling multicollinearity. Highly correlated or near-duplicate variables are common in biomedical tables. Detecting and managing redundancy helps reduce overfitting and makes the model’s reasoning easier to interpret because it becomes clearer which variables provide distinct information.

Resilience in HDLSS settings. The framework needs to work when the feature count is larger than the number of patients. That typically requires dimensionality-aware strategies that prevent the model from treating noise as signal.

Independence from feature names. The framework should not rely on semantic column identifiers. It should learn from distributions, relationships, and structure in the values themselves so it still works when feature names are anonymized or uninformative.

Strong overfitting control. Regularization, sparsity constraints, and structured priors can reduce the chance that the model memorizes artifacts specific to a dataset, improving the odds that it will generalize to new cohorts.

Modeling non-linear biological interactions. Biomedical outcomes often depend on non-additive effects and higher-order interactions across pathways. The framework should capture these non-linear dependencies rather than assuming purely linear relationships.

Compute efficiency and deployment practicality. The framework should aim for efficient training and inference. Using pretraining where appropriate, streamlined feature selection, and optimized inference can keep the compute burden low enough for practical use.

Overall, these capabilities define a framework that can handle HDLSS biomedical tables

while keeping rare signals visible, reducing sensitivity to correlated inputs and anonymized feature names, and producing predictions that are more stable, interpretable, and feasible to deploy.

3 Work Plan and Timeline

The project ran for twelve weeks and followed a planned sequence of tasks. The work was broken into phases so the team could finish one part, review it, and then move on without rushing later stages.

Work was split across team members based on interest and technical strengths. The group met regularly to check progress, raise issues early, and decide what to do next. Meetings with the supervisor were also part of the routine, mainly for progress tracking and clearing doubts. The team also completed three milestones during the same twelve-week period to present updates at different points in the project.

Project phases and who worked on what

3.1 Weeks 1–2: Literature review and dataset selection

The first two weeks focused on reading papers on tabular learning, machine learning methods, and healthcare prediction. Chethan, Bhavesh, Sourav, and Sajith reviewed existing approaches, including GRACES, DeepFS, XGBoost, and TabPFN. Sajith took the lead on dataset selection and handled the initial exploration work.

3.2 Weeks 3–4: Preprocessing and simulator work

After the datasets were chosen, the next step was preparing them for experiments. Chethan and Sajith worked on cleaning, preprocessing, and formatting. In parallel, Sajith and Sourav built and tested the simulator using SMOTE and SMOGN. The simulator went through several trials so the generated datasets could be checked before they were used for training.

3.3 Weeks 5–7: Embedding and feature-selection setup

During this phase, the focus moved to embeddings and feature selection. Sourav worked on implementing TabPFN. Sajith and Chethan worked on DeepFS and GRACES. The group tried different parameter settings during this period to improve embedding quality and stability.

3.4 Weeks 8–9: Embedding generation and model development

Once the embedding approach was settled, the team generated the embedded features and connected them with deep learning methods. Access to the UNITI dataset happened during this stage, but only after administrative steps were completed and an NDA was signed. Bhavesh and Sourav concentrated on generating and analyzing embeddings, while Sajith and Chethan built the classification and regression models. TabPFN was also tested in this phase and compared with the other methods being used.

3.5 Weeks 9–11: Training runs and evaluation

Most experiments and result collection happened here. Chethan and Sajith ran larger training jobs using DeepFS- and GRACES-based embeddings with XGBoost, along with TabPFN models. Bhavesh and Sourav focused on evaluation, using multiple metrics and checking stability under different conditions.

3.6 Weeks 11–12: Integration and final write-up

In the last stage, the components were brought together into a single experimental pipeline. Everyone contributed to testing, reviewing results, and preparing the final report and presentation.

Communication stayed consistent across the twelve weeks through email, meetings, shared repositories, and online collaboration tools. Supervisor meetings continued through the main phases for technical advice, progress checks, and clearing questions. Sourav led several progress presentations to track milestones and identify areas that needed correction.

Some of these discussions led to task reshuffling, scope adjustments, and extra attention to edge cases. The formal milestone presentations also gave the team a chance to collect feedback and update the methodology. When technical issues or time constraints came up, tasks were adjusted so the work could stay on schedule and finish within the twelve-week window.

3.7 Use of Gantt Chart for Project Management

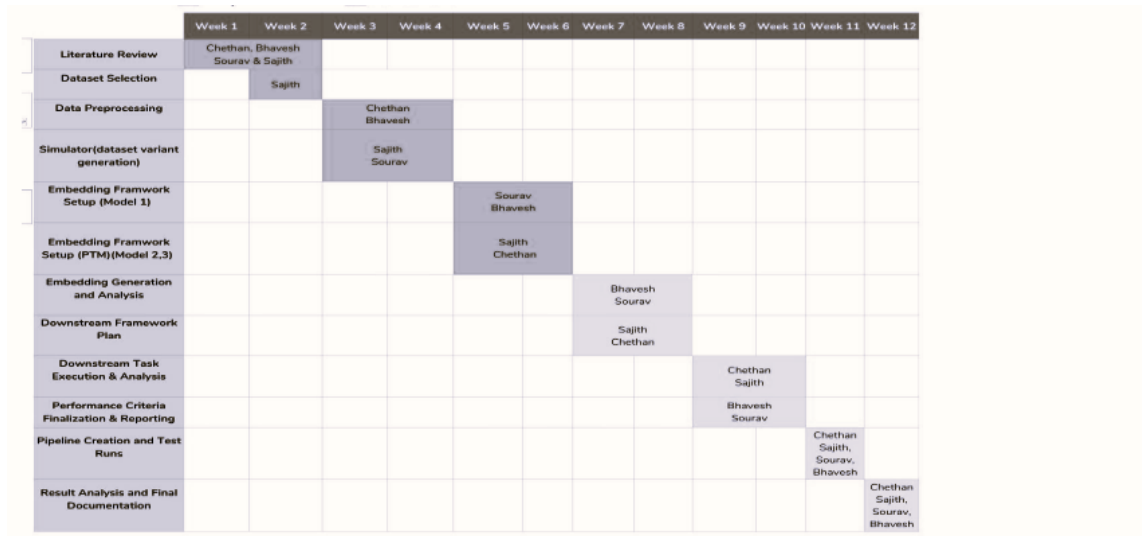


Figure 1: Gantt chart used to support project planning and monitoring

To support effective project planning and monitoring, a Gantt chart was used as a visual tool during the development process. The chart was first prepared and presented during the second project milestone to illustrate task distribution, timelines, and individual responsibilities. It helped the team track progress, identify delays, and adjust schedules when necessary.

By clearly mapping project phases to weekly time slots, the Gantt chart supported better coordination among team members and improved time management. It also served as a reference during supervisor meetings and milestone presentations to demonstrate work completed and upcoming objectives.

4 Data Description

Our study evaluated the models using three datasets. The first is a public drug-induced autoimmunity (DIA) dataset from the UCI Machine Learning Repository. The second is the UNITI Tinnitus dataset, which was provided only for auditory research. The third is a subset of the UNITI data that adds genetic information so we can analyze molecular/genomic signals alongside the clinical variables.

4.0.1 Drug-Induced Autoimmunity (DIA) dataset

The DIA dataset, curated by Huang (2025), includes molecular descriptors generated with the RDKit library for drug-induced autoimmunity research. It is set up for binary classification and contains 477 drug compounds. Each compound is represented by 195 numerical features that describe molecular and structural properties, and the dataset also includes categorical identifiers such as SMILES strings.

The data comes with predefined training and test splits and has no missing values. The target is a binary label indicating whether a compound is DIA-positive (1) or DIA-negative (0).

Dataset characteristics

- Type: Tabular
- Subject area: Health and medicine
- Task: Classification
- Instances: 477
- Features: 195
- Feature types: Continuous molecular descriptors, categorical identifiers
- Missing values: None
- Target variable: Label (1 = DIA-positive, 0 = DIA-negative)

How it is used

This dataset is used to train predictive models that flag potential autoimmune risk in drug candidates. Prior work, including the InterDIA framework, has used ensemble methods, feature selection, and SHAP-based interpretability and reported AUC values above 0.88 in cross-validation and external validation. Those results are used to connect model predictions back to molecular properties that may be linked to drug-induced autoimmunity and to support safer drug development.

Access and citation

The dataset is available through the UCI Machine Learning Repository under the Creative Commons Attribution 4.0 (CC BY 4.0) license. It includes training and test sets and descriptor metadata to support reproducible experiments.

4.0.2 UNITI Tinnitus RCT dataset

The UNITI Tinnitus RCT dataset contains patient-level data from a randomized controlled clinical trial. It includes baseline visit details, demographics, treatment assignments, and a range of clinical and psychological measures, including Tinnitus Handicap Inventory (THI) scores and questionnaire responses.

It is tabular and mixes continuous, categorical, and binary variables, with many fields stored as strings. Missingness is substantial in several columns, and some features are entirely missing across participants. The prediction target is the THI score at the final visit.

Dataset characteristics

- Type: Tabular
- Subject area: Health and medicine
- Task: Regression
- Instances: 250
- Features: 625
- Feature types: Continuous, binary, categorical, and text-based attributes
- Missing values: Present in multiple columns
- Target variable: Final visit THI score (0–100)

How it is used

In this study, baseline visit information is used to predict the final visit THI score. This design tests whether treatment outcomes can be estimated from initial patient characteristics, which can help clinicians form early expectations about likely outcomes and tailor interventions.

Access and citation

This dataset is not public. It was shared only with the research team under a Non-Disclosure and Confidentiality Agreement (NDCA), and it cannot be redistributed. Use of the dataset follows the NDCA terms and institutional ethical requirements.

4.0.3 UNITI Tinnitus dataset with genetic information

A third version of the UNITI dataset adds genetic information to the usual clinical and questionnaire variables. It includes two additional genetic features for about 250 patients, while those same attributes are missing for the remaining participants.

Because the genetic variables are only available for part of the cohort, the dataset has additional heterogeneity and missingness. The genetic fields were merged with baseline and clinical data before preprocessing and handled separately to reduce bias that could come from incomplete coverage.

Dataset characteristics

- Type: Tabular
- Subject area: Health and medicine

- Task: Regression
- Instances: 250
- Features: 627 (including genetic attributes)
- Feature types: Continuous, binary, categorical, and genetic markers
- Missing values: Present in genetic and clinical attributes
- Target variable: Final visit THI score (0–100)

How it is used

Adding genetic information makes it possible to test whether the genetic markers relate to tinnitus severity and whether they improve prediction when combined with clinical and questionnaire features. It also provides a practical setting for studying models when some features are only available for a subset of patients.

Access and citation

Access to the genetics-augmented dataset follows the same confidentiality rules as the original UNITI dataset. All analysis is carried out under the relevant ethical and legal requirements for handling sensitive patient data.

4.1 UNITI Tinnitus RCT Dataset

The UNITI Tinnitus RCT dataset contains patient-level data collected during a randomized controlled clinical trial. It includes baseline visit information, demographic characteristics, treatment assignments, and multiple clinical and psychological measures, such as Tinnitus Handicap Inventory scores and questionnaire responses.

The dataset is organized in a tabular format and contains a mixture of continuous, categorical, and binary variables, many of which are encoded as string objects. Several columns exhibit substantial missing values, with some features entirely missing for all participants. The primary prediction target is the final visit THI score.

Dataset Characteristics

- Type: Tabular
- Subject Area: Health and Medicine
- Task: Regression
- Number of Instances: 250
- Number of Features: 625
- Feature Types: Continuous, binary, categorical, and text-based attributes
- Missing Values: Present in multiple columns
- Target Variable: Final visit THI score (range: 0–100)

Dataset Use and Significance In this study, baseline visit data is used to predict the final visit THI score, enabling evaluation of treatment effectiveness based solely on initial patient characteristics. This setting provides clinicians with early insights into potential treatment outcomes and supports personalized intervention strategies.

Access and Citation This dataset is private and was shared exclusively with the research team under a Non-Disclosure and Confidentiality Agreement (NDCA). Due to confidentiality restrictions, public access is not permitted. Any use of the dataset complies with the terms of the NDCA and institutional ethical guidelines.

4.2 UNITI Tinnitus Dataset with Genetic Information

A third version of the UNITI dataset extends the standard clinical and questionnaire data by incorporating genetic information. This extended dataset includes two additional genetic features available for approximately 250 patients, while such information is missing for the remaining participants.

The partial availability of genetic attributes introduces additional heterogeneity and missingness, increasing the complexity of data integration and modeling. These features were merged with baseline and clinical variables prior to preprocessing and were treated separately to minimize bias introduced by incomplete coverage.

Dataset Characteristics

- Type: Tabular
- Subject Area: Health and Medicine
- Task: Regression
- Number of Instances: 250
- Number of Features: 627 (including genetic attributes)
- Feature Types: Continuous, binary, categorical, and genetic markers
- Missing Values: Present in genetic and clinical attributes
- Target Variable: Final visit THI score (range: 0–100)

Dataset Use and Significance The inclusion of genetic information enables investigation of potential associations between genetic markers and tinnitus severity. This extended dataset facilitates the evaluation of whether genetic factors contribute to improved prediction performance when combined with clinical and questionnaire-based features. At the same time, it provides a realistic setting for studying learning under partial feature availability in medical data.

Access and Citation Access to the genetic-augmented dataset is governed by the same confidentiality agreements as the original UNITI dataset. All analyses were conducted in compliance with ethical and legal requirements for handling sensitive patient information.

5 Methodology

This section outlines the experimental framework used in this study, covering data preprocessing, simulation of imperfect data conditions, model training, and performance evaluation. The goal is to make results reproducible, keep model comparisons fair, and obtain a reliable picture of performance on high-dimensional healthcare datasets.

5.1 Data Preprocessing

Every dataset that was included in the current research project went through a preprocessing phase prior to utilizing these datasets for model training and evaluation. As there were three datasets that were utilized in the current research project, common preprocessing steps were applied consistently across the three datasets to allow for comparability and consistency in outcomes. The common preprocessing steps that were taken cared for general data quality issues and prepared the datasets for downstream analysis. In addition to the common preprocessing steps taken, each dataset required its own separate data preprocessing procedures that were specific to the way in which the dataset was structured, composed of features, and the distribution of the data itself.

Additionally, preprocessing was not received at just the dataset level; there were also different preprocessing strategies taken per dataset due to the specific machine learning algorithms that would be used for training and testing each dataset, since these algorithms have their own unique input requirements and sensitivity to feature scaling, encoding, and sampling methods used within them. The complete list of preprocessing procedures (common and specific to dataset and algorithm) are defined in the following subsections.

Common Preprocessing Steps: Any column/feature with greater than 95% of the values missing in that column/feature was excluded as it is unlikely that a sufficient amount of the data would yield any reliably usable signal for the model. Also excluded from consideration were columns/features that had identifier-level characteristics. These serve essentially as column/feature name or label for individual records in the dataset and provide no information useful for predictive modeling. Any feature/column that had data that was consistent across all records in the particular column was also excluded from consideration, as there is no variability and therefore no separation within the feature/column from the rest of columns/features in the dataset. Also, duplicate values were removed as well.

Drug-Induced Autoimmunity dataset Duplicate entries were eliminated from the Drug-Induced Autoimmunity dataset and feature types were explored to find continuous, integer, binary, and categorical features. Retaining integer and binary variables in their original structure was done to ensure that they could easily be understood, whereas categorical identifiers such as SMILES strings were not subjected to transformation and only retained as identifiers. A few steps in the dataset pre-processing was slightly different for each algorithm as described below:

- For TabPFN, no explicit feature normalization was applied, as the model internally performs feature scaling during training.
- The similarity-based approaches could require further pre-processing due to specific algorithms that are being used on those datasets. Continuous molecular descriptors should be standardised (zero mean and unit variance), because these methods can be affected by

scale. In addition, a missing value check was performed. Missing values would have to be fixed before applying similarity-based methods, since similarity-based methods cannot handle missing data directly.

UNITI Tinnitus datasets (No genetic information): All records corresponding to the final visit information were removed from this dataset to ensure that all of the inputted data consisted only of baseline values. However, the THI score of the final visit was kept and added as a separate outcome feature. In addition, a separate categorical variable was added to quantify THI severity for the baseline THI score, by having scores divided into five clinically meaningful categories: Slightly (0–16), Mildly (18–36), Moderately (38–56), Severely (58–76), and Catastrophically (78–100). Unfortunately, all features in the dataset were originally stored as strings. Therefore, all numerical feature values were converted from string format into their appropriate numeric types in order to allow for valid mathematical processes to be performed on them and for training of models.

- For TabPFN, no explicit feature normalization was applied, as the model internally performs feature scaling during training. Also no missing values were removed at this stage since this model can handle them internally.
- The similarity-based approaches could require further pre-processing due to specific algorithms that are being used on those datasets. Also, all the features with string data or categorical data was encoded into numbers as these models can't work on such values but only on numerical data. Continuous molecular descriptors should be standardised (zero mean and unit variance), because these methods can be affected by scale. In addition, a missing value check was performed. Missing values would have to be fixed before applying similarity-based methods, since similarity-based methods cannot handle missing data directly. Upon encountering missing values, we have chosen to fill them with value of -1.

UNITI Tinnitus datasets (With genetic information): The preprocessing for this version of the dataset is similar to that performed on the previous variant of this dataset without genetic information except for the addition of genetic features. These were stored in a separate dataset file and were integrated into this dataset file by matching records between them based on the patient's external identifier columns common to both datasets.

Finally, we have also kept track of all the features we have removed from the datasets as a part of preprocessing in case they were relevant later on.

5.2 Simulation of Data Imperfections

In order to assess predictive ability under realistic and difficult data conditions, we developed a simulator module for injecting controlled imperfections into the experimental datasets. The simulator replicates two major problems typically encountered during clinical and biomedical prediction tasks: limited sample size and imbalanced target distributions. This framework will allow us to perform structured evaluations of model robustness against adverse data scenarios in both classification and regression environments.

Sample Size Reduction. To emulate data scarcity in predictive environments, a controlled sample size reduction procedure was applied. Given an original dataset (X, y) containing N samples, a subset of size $N' = \lfloor rN \rfloor$ was selected using randomized sampling, where $r \in (0, 1]$ denotes the retention ratio. The reduced dataset (X_r, y_r) was used for subsequent simulation steps. This procedure reflects practical constraints such as limited patient recruitment, restricted data availability, and privacy regulations.

Class Imbalance in Classification and SMOTE. In classification-based healthcare prediction tasks, class distributions are often highly imbalanced, with minority classes corresponding to rare but clinically important outcomes. Models trained on such data tend to exhibit biased predictions toward majority classes, resulting in poor sensitivity for minority cases.

To address this issue, the simulator incorporates an oversampling strategy based on the Synthetic Minority Over-sampling Technique (SMOTE).¹³ SMOTE generates synthetic minority-class samples by interpolating between neighboring instances in feature space. For two minority samples x_i and x_j , a synthetic instance is generated as

$$x_{\text{new}} = x_i + \alpha(x_j - x_i),$$

where $\alpha \sim \mathcal{U}(0, 1)$. This approach increases minority class representation without duplicating existing samples, thereby reducing overfitting and improving class discrimination.

Target Imbalance in Regression and SMOGN. In regression-based prediction tasks, target variables often exhibit skewed or imbalanced distributions, where extreme or clinically relevant values are sparsely represented. Learning algorithms trained on such data tend to bias predictions toward dominant regions of the target space.

To address this challenge, an oversampling strategy inspired by the SMOTE for Regression with Gaussian Noise (SMOGR) method¹⁴ was applied. In the implemented SMOGR-like procedure, rare target regions were identified using percentile-based thresholds. Let q_{25} and q_{75} denote the 25th and 75th percentiles of the reduced target distribution y_r . Samples satisfying

$$y_r \leq q_{25} \quad \text{or} \quad y_r \geq q_{75}$$

were classified as rare instances.

For each synthetic sample, two rare instances (x_i, y_i) and (x_j, y_j) were randomly selected. A new synthetic instance was generated through linear interpolation:

$$x_{\text{new}} = x_i + \alpha(x_j - x_i), \quad y_{\text{new}} = y_i + \alpha(y_j - y_i),$$

where $\alpha \sim \mathcal{U}(0, 1)$. This procedure increases the representation of under-sampled target regions while preserving local data structure.

Integrated Simulation Procedure. The complete simulator operates in two sequential stages. First, the dataset is reduced according to the specified retention ratio. Second, imbalance correction is applied using SMOTE for classification tasks and SMOGR for regression tasks. The final simulated dataset (X_s, y_s) is constructed by combining original and synthetic samples. Descriptive statistics, including target range, mean, and standard deviation, are computed before and after simulation to monitor distributional changes.

Application in Predictive Evaluation. The simulated datasets were used to evaluate prediction performance under varying degrees of data scarcity and imbalance. By adjusting the retention ratio and oversampling intensity, multiple experimental scenarios were generated for both classification and regression tasks. These scenarios enabled systematic comparison of embedding-based, ensemble-based, and foundation-model approaches under controlled perturbations in realistic clinical and medical prediction settings.

5.3 Model Algorithms

This section describes the learning frameworks employed for downstream prediction, including embedding-based methods combined with ensemble models and foundation-model approaches. Each configuration is designed to evaluate different representation learning and inference paradigms for high-dimensional tabular healthcare data.

5.3.1 Deep Feature Screening (DeepFS)

Deep Feature Screening (DeepFS) It is basically a feature selection algorithm that selects the most useful features for prediction using an autoencoder-based approach. In DeepFS, we send in the dataset, which is then fed into an autoencoder.¹¹ This autoencoder will transform the data into a low-dimensional representation of itself, much like creating a summary of the original data. Then it compares the original feature list to the low-dimensional representation and determines which are the most informative features that will help for prediction. It does this comparison by computing the rank distance correlation for each feature. It assigns a score for each feature based on this, and then based on this score we make a ranking of the most important features. The top features are then selected for the output either by using the jump method or based on a number of features specified by the user. It combines autoencoder-based dimensionality reduction with multivariate rank distance correlation to identify informative and non-redundant features. The method is model-free and distribution-free, enabling it to capture nonlinear feature interactions in both supervised and unsupervised learning scenarios.

The selected features will then be provided to a downstream predictive model for prediction purposes. In our approach, we will be performing either a classification or regression task using XGBoost and TabPFN with the filtered data set. Therefore, the combination of DeepFS with these models will result in improved predictive accuracy and stability of the model, as well as reducing the noise inherent in using all raw features and the computational complexity associated with doing so.

Raw Data → Preprocessing → DeepFS Feature Selection → Downstream Model →
Prediction

5.3.2 GRACES (GRaPh Convolutional nEtwork feature Selector)

It is basically another deep learning-based feature selector which uses latent relationships between samples to do feature selection.¹² It keeps a bias feature as the initial member of the selected set and iteratively selects one feature at a time. In each iteration, a similarity graph is constructed using only the features currently selected, with nodes representing samples and edges determined by cosine similarity above a set threshold. It has a three-layer neural network, with an input linear layer, a GraphSAGE GCN layer, and an output layer. These combinations compute feature scores while ensuring that only the candidate features are evaluated. To reduce any overfitting, GRACES uses multiple dropouts to average gradients, injects Gaussian noise

into GCN weights, and applies an ANOVA F-test to combine statistical and gradient-based scores. Each candidate feature receives a combined score, and the feature with the highest score is added to the selected set, repeating until the desired number of features is chosen. Through the clustering assumption that similar samples likely belong to the same class, it identifies informative features that capture the underlying structure of HDLSS datasets.

The selected features will then be provided to a downstream predictive model for prediction purposes. In our approach, we will be performing either a classification or regression task using XGBoost and TabPFN with the filtered data set. Therefore, the combination of GRACES with these models will result in improved predictive accuracy and stability of the model, as well as reducing the noise inherent in using all raw features and the computational complexity associated with doing so.

Raw Data → Preprocessing → GRACE Embedding Generation → Downstream Model →
Prediction

5.3.3 TabPFN (Tabular Prior-data Fitted Network)

It is basically a tabular foundation model designed to perform supervised learning tasks, such as classification and regression, in a single neural network forward pass.⁶ It employs in-context learning which was inspired by large language models (LLMs), to learn a tabular prediction algorithm directly from data. After this, it approximates Bayesian inference using a prior defined through synthetic datasets. It is capable of handling missing values on its own. The workflow consists of three stages, first the synthetic data generation, then the pre-training part, and then the inference. During pre-training, the model predicts masked targets while using unmasked samples as context. TabPFN performs inference with both labelled training data and unlabelled test data together, generating predictions for both datasets at once in one forward pass, with no dataset-specific training, no gradient updates, and no hyperparameter tuning. In terms of architecture, TabPFN uses a two-way attention mechanism that attends to features in each row and features in each column to ensure invariance to both the ordering of samples and features. In addition, it supports caching of the state of the training context for efficient reuse and provides support for missing values through an additional input bit, maps categorical features to integers, z-normalizes numerical features, and outputs probabilistic predictions for uncertainty modelling and density estimation.

In our specific case we use the TabPFN directly to make predictions based on the input. As TabPFN handles everything natively it requires minimum preprocessing. One thing to be noted here is that it is possible to extract the embeddings from this model to see how the data is represented as vectors inside the model. It is also possible to pass the embeddings to other downstream models and create hybrid models that makes use of TabPFN generated embeddings to make predictions. However in our project, when we tried to follow this approach we had to face significant drop in the quality of the predictions. Hence we finalised to make the model purely TabPFN and not to use it as a hybrid since we found it offered better results.

5.4 Evaluation Metrics

The performance of the model was assessed based on task-specific metrics, which were chosen based on the nature of the problem for the prediction task. Different methods of assessment were used for the regression task and the classification task

5.4.1 Evaluation Metrics for Classification (DIA Dataset).

The issue of class imbalance in the "Drug-Induced Autoimmunity (DIA)" dataset is determined to be quite prominent even before the oversampling techniques are used. This means that the number of instances in the positive class for "DIA" is quite small. In this scenario, the accuracy metric may not be very effective, as it may be possible to attain a high accuracy rate by simply predicting the majority class. To avoid this issue, the weight F1 score was used as the main evaluation metric for the performance of the model even before the oversampling techniques were used. The weighted F1-score aggregates class-wise F1-scores according to class proportions, thereby accounting for imbalance and providing a balanced measure of classification performance

After applying SMOTE-based oversampling, class distributions became more balanced. Under these conditions, accuracy and standard F1-score were adopted as evaluation metrics.

Where F1 score is given by

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}},$$

where precision and recall are given by

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

5.4.2 Evaluation Metrics for Regression (UNITI Dataset).

The UNITI Tinnitus dataset is a regression task where the prediction of the final visit Tinnitus Handicap Inventory (THI) score is required, which varies between 0 and 100. The evaluation metrics used were Mean Absolute Error, Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

The Mean Absolute Error measures the average magnitude of prediction errors and is defined as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|.$$

The Root Mean Squared Error is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}.$$

The coefficient of determination is defined as

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}.$$

Interpretation in the Context of THI Score Range As the range of THI scores is $[0, 100]$, MAE and RMSE are clinically interpretable. For example, if $\text{MAE} = 5$, it means the average deviation in the predictions is five units of the THI scale. Lower MAE and RMSE values imply better estimation of tinnitus severity, while higher R^2 values indicate stronger explanatory power of the predictive model.

6 Unified Pipeline

We built a single experimental pipeline so every method is run under the same procedure and can be compared under identical settings. The pipeline fixes the sequence of steps—from generating dataset variants, to constructing representations, to training models, to evaluating results. Even though the order is fixed, the stages are still independent, so we can change one stage (for example, swap a feature selection method) without rewriting the rest of the workflow.

Dataset preprocessing stays outside the pipeline. Each dataset needs its own preprocessing because the structure, feature formats, and data quality issues differ. Keeping preprocessing separate makes those dataset-specific decisions explicit, while everything after preprocessing follows the same procedure across datasets.

6.1 Pipeline architecture

Initially, the pipeline will have an input dataset that will either be divided into training and testing datasets or input directly into the respective modules to be processed and produce valid prediction outcomes. Once the input data has been processed and split correctly, the simulation module will utilize the training dataset to remove samples to simulate a limited number of samples (a scarcity of samples). The simulation module can apply synthetic oversampling to create balance in the occurrence of rare targets by creating (synthetic) samples of targets that do not have enough actual observations in the training sample.

After the simulation module has simulated the training dataset, the feature selection and representation modules will operate on the resultant training dataset to isolate the most useful features in order to provide a compressed dataset (a reduced number of dimensions) of the original dataset and/or compressed representation of the training dataset) that will provide a simplified, high signal input to the model for training, increasing model learning performance and aligning on relevant model inputs (model training).

Finally, as the completed training dataset will serve as input to the final prediction module(s) for producing predictions using the input training dataset(s). Also, a performance evaluation module will be needed to compare the performance result with the original prediction criteria used with the original training sample. By using all of these modules and the associated data conversion functions the pipeline will systematically evaluate all possible experimental and data sampling conditions and all possible features that can be used for model development for both regression and classification and provide a systematic and reproducible analysis of model predictions.

6.2 Data simulation module

The data simulation module applies controlled changes to the preprocessed datasets. It reduces the effective sample size and handles imbalance using oversampling methods based on SMOTE and SMOGN. The output is a set of dataset variants that reflect common constraints in clinical datasets, such as small sample sizes and skewed target distributions. We pass these variants to the remaining modules so every learning method is tested under the same simulated conditions.

6.3 Feature selection and representation learning module

This module reduces the size of very high-dimensional inputs by producing more compact representations. We run DeepFS for feature selection in ultra high-dimensional settings, focusing

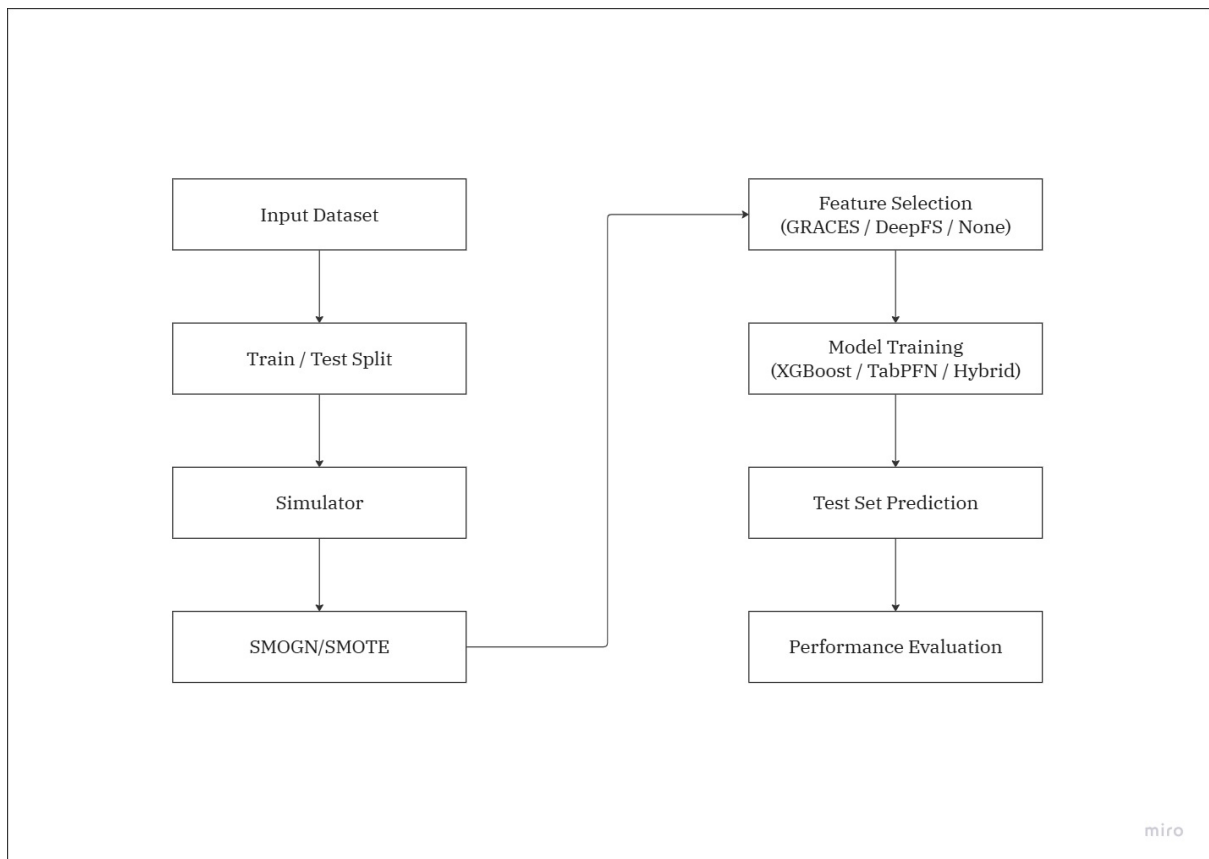


Figure 2: Pipeline architecture showing the workflow of the system

on attributes that carry predictive signal. In parallel, we use GRACES to produce feature set that capture nonlinear interactions and heterogeneous clinical patterns.

Together, the selected feature subsets reduce redundancy and limit the effect of noise present in the original feature space. They also provide structured inputs that downstream models can use directly.

6.4 Downstream prediction module

The downstream prediction module trains supervised models on the selected features and learned representations. For classification tasks, we train gradient-boosted decision trees to identify disease-related patterns and risk indicators. For regression tasks, we train gradient-boosted regressors to predict continuous outcomes, including THI scores.

We also include Tabular Prior-Fitted Network (TabPFN) as a foundation-model baseline for both classification and regression. In this setup, TabPFN uses in-context learning on raw or minimally processed feature values rather than relying on explicit feature engineering.

Training includes hyperparameter tuning and cross-validation. We keep the training procedure consistent across configurations so that differences in results reflect the learning method rather than changes in the training setup.

We include three model families to cover complementary inductive biases. As a tree-based benchmark, we use XGBoost with a compact configuration to reduce overfitting on small cohorts while retaining nonlinear decision boundaries. As a graph-based method, GRACES constructs a k -NN patient similarity graph and applies graph convolutions so that patients with similar outcomes can reinforce shared signal through their neighbours. As a manifold-based method, DeepFS relies on deep autoencoder-style compression to filter noise and highlight features that are essential for reconstructing the underlying data geometry. Finally, TabPFN acts as a transformer-based foundation model that performs in-context learning on (mostly) raw features, and we also evaluate two hybrids: GRACES_TABPFN, which feeds graph-informed representations into TabPFN, and DeepFS_TABPFN, which combines denoised feature subsets with TabPFN for prediction.

6.5 Evaluation and reporting module

The evaluation module computes task-appropriate metrics: weighted and standard F1, accuracy, MAE, RMSE, and R^2 . We aggregate results across dataset variants and simulation settings to measure how stable each method is when data conditions change.

For reporting, the module produces summary tables and visualizations and runs statistical comparisons to help interpret differences across outcomes.

6.6 Reproducibility and modularity

Reproducibility and modularity are design requirements. Each module is implemented as an independent component with fixed input and output formats. We record random seeds, configuration parameters, and experimental settings so the full set of runs can be replicated.

Because modules communicate through stable interfaces, we can update or replace individual components without restructuring the overall pipeline, which keeps the workflow maintainable as methods change.

7 Experimental Setup

7.1 Hyperparameters

This section shows the hyper params that were used in this pipeline for different models. All testing was performed with these hyperparameter values. Table: 1 explains the values for each hyper param per model used for this experimental setup. Meanwhile Table: 2 shows what each hyperparameter mentioned here means.

Table 1: Hyperparameters Used per Model in the Regression Pipeline

Model / Component	Hyperparameter	Value
XGBoost	n_estimators	1000
	max_depth	4
	learning_rate	0.01
	subsample	0.8
	random_state	42
TabPFN	n_estimators	32
	random_state	42
GRACES	n_features	50
	hidden_size	64, 32
	epochs	100
	batch_size	8
	alpha	0.95
	learning_rate	0.001
DeepFS	n_features	50
	latent_dim	15
	lr	0.001
	epochs	500
	lambda_reg	0.01

Table 2: Hyperparameter Definitions

Hyperparameter	Definition / Description
n_estimators	Number of boosting rounds (XGBoost) or transformer ensembles (TabPFN).
max_depth	Maximum depth of trees in XGBoost. Controls complexity; higher = more overfit.
learning_rate	Step size shrinkage in XGBoost / optimizer learning rate.
subsample	Fraction of training data used per XGBoost tree. Reduces overfitting.
random_state	Seed for reproducibility.
hidden_size	GRACES GNN hidden layer sizes, e.g., [64,32].
epochs	Number of training iterations (DeepFS or GRACES).
batch_size	Number of samples per batch in GRACES GNN training.
alpha	Quantile threshold in GRACES for graph edge sparsity.
n_features	Number of features to select in GRACES / DeepFS.
latent_dim	Size of latent space in DeepFS autoencoder.
lr	Learning rate for DeepFS optimizer.
lambda_reg	Weight for reconstruction loss in DeepFS autoencoder.

7.2 SMOGN Oversampling

SMOGN is used to handle imbalanced regression targets by oversampling under-represented regions of the target distribution. Here the oversampling cap used is 100%

7.3 Random Seeds & Reproducibility

Fixed random seeds ensure that all experiments can be reproduced with identical results across runs. Here the value used is 42 to ensure reproducibility.

7.4 Simulator Data Reduction Levels

The simulator was tested with different data reduction levels to evaluate model performance under varying feature availability. The table below shows each reduction percentage and the corresponding amount of data retained.

Table 3: Data Reduction Levels in the Simulator Module

Reduction Percentage	Data Retained (%)
0.0	100%
0.3	70%
0.6	40%

8 Results

8.1 UNITI Tinnitus datasets (No genetic information)

This experiment evaluates predictive performance on a Uniti Tinnitus non-genetic dataset. By excluding genetic variables, the analysis isolates the interaction between feature selection methods (DEEPFS, GRACES) and downstream learners (XGBoost, TABPFN). Model performance is evaluated using R^2 , RMSE, and MAE under varying data availability conditions, with and without SMOGN-based synthetic oversampling.

8.1.1 Scenario 1: 100% Data

The scenario where we have complete access to a large dataset is referred to as a “data-rich” setting; therefore, we can evaluate the best-possible performance for each model (i.e., without being victims of small-data effects). This scenario introduces questions of compatibility between different types of feature selection methods and machine learning algorithms. For example, there are clear significant differences in results between TABPFN-based models that do not utilize synthetic oversampling, in comparison to those models that utilize XGBoost-based pipelines that do incorporate SMOGN to create more robust models when compared to the former.

Table 4: Scenario 1 (100% Data): Pure Data without SMOGN

Method	R^2	RMSE	MAE
GRACES + TABPFN	0.4656	14.4944	11.4469
TABPFN (No FS)	0.4638	14.5183	11.4961
XGBoost (No FS)	0.4375	14.8713	11.7745
DEEPFS + XGBoost	0.4255	15.0282	11.7516
DEEPFS + TABPFN	0.4122	15.2010	11.7266
GRACES + XGBoost	0.3221	16.3245	12.6081

Table 5: Scenario 1 (100% Data): With SMOGN

Method	R^2	RMSE	MAE
DEEPFS + XGBoost	0.4330	14.9297	11.9130
XGBoost (No FS)	0.4285	14.9887	11.9531
DEEPFS + TABPFN	0.4285	14.9899	11.7916
TABPFN (No FS)	0.4153	15.1613	12.2043
GRACES + XGBoost	0.3835	15.5685	12.5510
GRACES + TABPFN	0.3678	15.7656	12.2291

8.1.2 Scenario 2: 70% Data

The evaluation of this scenario examines model performance under moderate reduction in data, simulating real-world situations in which only 70% of the total data is present. The differences in model performance in these circumstances highlight how well feature selection can overcome sample size limitations. The results yield the following conclusions: 1) DEEPFS using TABPFN achieves the best performance among the configurations tested despite the lack of SMOGN, and 2) DEEPFS when combined with XGBoost benefits consistently from synthetic oversampling; thus DEEPFS paired with either TABPFN or XGBoost will provide improved performance with regard to different sample sizes.

Table 6: Scenario 2 (70% Data): Pure Data without SMOGN

Method	R^2	RMSE	MAE
DEEPFS + TABPFN	0.5104	13.8741	10.7851
TABPFN (No FS)	0.4683	14.4582	11.2646
GRACES + TABPFN	0.4483	14.7269	11.8472
XGBoost (No FS)	0.4051	15.2934	12.1988
GRACES + XGBoost	0.3849	15.5501	12.1664
DEEPFS + XGBoost	0.3715	15.7196	12.1349

Table 7: Scenario 2 (70% Data): With SMOGN

Method	R^2	RMSE	MAE
DEEPFS + XGBoost	0.4527	14.6679	11.4928
DEEPFS + TABPFN	0.4232	15.0589	11.1443
TABPFN (No FS)	0.4013	15.3419	12.5113
XGBoost (No FS)	0.3997	15.3629	12.3731
GRACES + TABPFN	0.2794	16.8320	13.1756
GRACES + XGBoost	0.2389	17.2978	13.9655

8.1.3 Scenario 3: 40% Data

The evaluation of this scenario places stress on the evaluation of models in the presence of extreme data scarcity, in this case only retaining 40% of the original dataset for testing. The purpose of this scenario is to examine the robustness and stability of the pipeline models given such extreme reductions in data size. The findings indicate that DEEPFS when combined with TABPFN is the most robust pipeline in these extreme testing scenarios, while many of the XGBoost configurations suffer greatly in performance or are unable to provide meaningful performance results. Among the GRACES models the negative effects of SMOGN were the strongest in this scenario.

Table 8: Scenario 3 (40% Data): Pure Data without SMOGN

Method	R^2	RMSE	MAE
DEEPFS + TABPFN	0.4947	14.0944	11.3926
TABPFN (No FS)	0.3813	15.5963	12.6786
XGBoost (No FS)	0.3813	15.5965	12.3413
GRACES + TABPFN	0.3648	15.8022	12.6581
GRACES + XGBoost	0.2205	17.5062	14.3975

Table 9: Scenario 3 (40% Data): With SMOGN

Method	R^2	RMSE	MAE
DEEPFS + TABPFN	0.4188	15.1158	11.9862
XGBoost (No FS)	0.3701	15.7359	12.4526
DEEPFS + XGBoost	0.3646	15.8047	12.6365
TABPFN (No FS)	0.2813	16.8090	13.9415
GRACES + XGBoost	0.1181	18.6204	14.5120
GRACES + TABPFN	0.1145	18.6585	15.2773

8.2 UNITI Tinnitus datasets (With genetic information)

The genetic attributes present greater challenges for tabular models than clinical ones because they are more than 10 times more diverse, include higher latent interaction (epistasis), and their overall signal-to-noise ratio is significantly lower. Consequently, it is difficult to use traditional approaches to tabular predictions. These factors influence our use of structure-informed feature learning algorithms. The performance of models is assessed using the appropriate R^2 , RMSE and MAE metrics under varying data availability conditions and using or not using SMOGN-based synthetic oversampling.

8.2.1 Scenario 1: 100% Data

In the fully-dense scenario, we evaluate model performance for the complete genetic feature space. This scenario is considered the highest signal level that any model will see, while providing models with extreme noise levels and the highest number of features available for input. The results show that in this case, graph-based feature learning methods outperform traditional tabular methods; and therefore, when using GRACES (the graph based method) in conjunction with TABPFN (a tabular method), the hybrid performs best, particularly in conjunction with the SMOGN-based method of generating synthetic features.

Table 10: Scenario 1 (100% Data): Pure Data without SMOGN

Method	R^2	RMSE	MAE
GRACES + TABPFN	0.1894	18.4066	13.8351
XGBoost (Baseline)	0.1647	18.6853	14.1034
DEEPFS + TABPFN	0.1510	18.8379	13.7218
DEEPFS + XGBoost	0.1247	19.1270	13.8374
GRACES + XGBoost	0.1137	19.2470	14.4465
TABPFN (No FS)	0.1027	19.3656	14.2458

Table 11: Scenario 1 (100% Data): With SMOGN

Method	R^2	RMSE	MAE
GRACES + TABPFN	0.2200	18.0561	13.3596
DEEPFS + TABPFN	0.1693	18.6333	13.2915
DEEPFS + XGBoost	0.1549	18.7936	13.9705
XGBoost (Baseline)	0.1439	18.9162	14.1638
GRACES + XGBoost	0.0712	19.7033	15.0403

8.2.2 Scenario 2: 70% Data

In the next scenario, we evaluate model performance under moderate genetic feature deletion. Approximately 30% of genetic features have been deleted. In comparison to prior scenarios, the predictive performance of the majority of models experienced decreased performance due to the sensitivity of genetic prediction when removing genetic features. While GRACES-based hybrids are still competitive when SMOGN is not used, the performance of DEEPFS based pipelines is enhanced when SMOGN is used, suggesting that the two approaches provide markedly different levels of robustness against loss of genetic features.

Table 12: Scenario 2 (70% Data): Pure Data without SMOGN

Method	R^2	RMSE	MAE
DEEPFS + XGBoost	0.1454	18.8995	14.3534
GRACES + TABPFN	0.1444	18.9109	14.7065
TABPFN (No FS)	0.1263	19.1096	13.9715
XGBoost (Baseline)	0.1080	19.3081	14.5154
DEEPFS + TABPFN	0.0943	19.4565	15.0681
GRACES + XGBoost	0.0534	19.8905	15.2991

Table 13: Scenario 2 (70% Data): With SMOGN

Method	R^2	RMSE	MAE
DEEPFS + XGBoost	0.1853	18.4530	13.4081
DEEPFS + TABPFN	0.1310	19.0583	14.5718
TABPFN (No FS)	0.1051	19.3400	14.0465
XGBoost (Baseline)	0.0992	19.4034	14.6606
GRACES + TABPFN	-0.0157	20.6041	14.8955
GRACES + XGBoost	-0.0224	20.6715	16.2009

8.3 Drug-Induced Autoimmunity Dataset

This section evaluates classification performance on the Drug-Induced Autoimmunity dataset under two experimental scenarios. The objective is to analyze the impact of simulator-driven balancing in combination with feature selection and downstream classifiers. Performance is reported using Accuracy, F1-score, and weighted F1-score to account for class imbalance.

8.3.1 Scenario 1: Classification Without Simulator

Training was carried out without either synthetic balancing or simulator-derived reduction of the original dataset to provide baseline performance and observe natural class imbalance. XGBoost (without feature selection) produced the highest F1 score indicating tree-based models are capable of exploiting strong marginal signals as long as there are enough raw features to work with. TabPFN-based models showed close to XGBoost performance through competitive accuracy and weighted F1 scores despite not being explicitly balanced providing further evidence that they are robust in low-sample conditions. The analyses demonstrate that feature selection methodologies using DeepFS and GRACES did not have a consistent impact on performance due to aggressive feature reduction likely eliminating weak, but collectively informative, signals needed for making accurate minority-class discrimination when there is no balance.

Table 14: Classification Performance Without Simulator

Feature Selection	Balancing	Classifier	Accuracy	F1	F1 (Weighted)
None (196 features)	None	XGBoost	0.7992	0.5000	0.7819
None (196 features)	Class Weights	XGBoost	0.7741	0.4808	0.7631
DeepFS (100)	None	TabPFN	0.7950	0.4096	0.7608
None	None	TabPFN	0.7950	0.4096	0.7608
GRACES (100)	None	TabPFN	0.7741	0.3415	0.7347

8.3.2 Scenario 2: Classification With Simulator

This situation assesses how simulating tabular data before it goes through the SMOTE process increases classifier accuracy and F1 score consistency than using the classifiers without the simulation. Compared to when the classifiers were not interfacing with a simulator, they were more accurate and had a more consistent F1 score than when the classifiers were used without interfacing with a simulator. Therefore, the simulators serve to regularize the process of learning by decreasing the number of times the classifiers have been given the same data; whereas,

synthetic oversampling helps out the minority class by increasing the number of times the classifiers have seen data representing the minority class. Moreover, the combination of DeepFS and TabPFN methods produced the most accurate results of all combinations; therefore, the pair of feature manifold-based feature selection with controlled synthetic oversampling works the best for predicting the outcome correctly. Ultimately, the results from the current study indicate that using a simulator as a part of a machine learning process stabilizes classification accuracy when dealing with imbalanced datasets that involve High Dimensional Low Sample Space dataset features.

Table 15: Classification Performance With Simulator (50% Reduction + SMOTE)

Feature Selection	Balancing	Classifier	Accuracy	F1	F1 (Weighted)
DeepFS (100)	Simulator (50% + SMOTE)	TabPFN	0.8075	0.4390	0.7740
GRACES (100)	Simulator (50% + SMOTE)	TabPFN	0.7908	0.4444	0.7658

9 Discussion

The findings discussed in this section highlight the main results from the experimental study and provide an analysis of the performance of several different modeling techniques across several datasets and experimental setups.

The results show that ensemble models like a combination of tabular pre-fitted models and robust feature selection techniques, consistently reach superior performance compared to traditional approaches, regardless of the dimensionality of the dataset or how limited the sample size is. In particular, the DeepFS and GRACES methodologies produced feature representations that, when combined with TabPFN, showed reliable and comparable performance for both classification and regression tasks. These methods demonstrated an ability to capture complex, non-linear feature interactions as well as reduce redundancy in the raw tabular datasets.

The experiments demonstrated that class imbalance and small sample sizes affect the behaviour of the models significantly across datasets, most notably the Drug-Induced Autoimmunity dataset. The implementation of SMOTE for classification and SMOGN for regression proved to assist with creating a more equal distribution of classes, which forces the model to pay attention to outlying value ranges which are extremely important in this healthcare domain. The implementation of SMOGN for oversampling in regression tasks contributed to improved stability of the predictions for rare target values in the UNITI dataset.

Although TabPFN showed good performance as a foundation model, although the performance varied across different datasets. Since TabPFN could also be applied as a classifier and regressor, it provided an important set of baseline comparisons. It gave greater results with combination of feature selectors. It clearly proved that it can give reliable predictions even when the samples were reduced

Another trend that was observed was the variability of synthetic oversampling dependent on data availability (balance) and feature type. For the non-genetic data in the UNITI system, SMOGN was not consistently providing performance improvements in areas of abundant data and in many instances reduced the accuracy of prediction; GRACES-based pipelines were affected most. This indicates that when enough real samples are available, the addition of synthetic data can introduce additional noise rather than provide a meaningful signal. In cases where there was a moderate or severe amount of data reduction, SMOGN provided measurable benefits to DEEPFS-based pipelines, indicating that manifold-based feature selection methods

are better able to accommodate the distributional data changes created by synthetic samples. Across genetic datasets, predictive performance was seen to decrease fairly quickly as features are lost, indicating that genetic prediction tasks are quite sensitive to feature loss. Even so, hybrid GRACES-based models were relatively robust across all data availability scenarios, reinforcing the potential of conducting structure-aware feature learning in high-dimensional low signal-to-noise settings. Finally, the results of the classification experiments demonstrate that the combination of simulator-driven data reduction and controlled oversampling improved the stability and consistency of performance measures at beyond maximizing peak accuracy, thereby supporting the presence of the simulator as a regularization component in the overall predictive pipeline.

The overall experimental framework used permitted a systematic comparison of the various methodologies.

9.1 Limitations

However, a few additional aspects has to be taken into account apart from the prediction quality.

Firstly, the dataset used for this study is limited in size, and hence this cannot be completely reliable in the case of datasets of bigger size or different degrees of instances to feature ratio. Another factor is that some datasets can be more heterogeneous in nature, and this work has to be extended to tailor such cases.

Secondly, since not all the patient records included their genetic data, there was more variability (heterogeneity) and missing data across patients which should have been appropriately addressed during preprocessing; however, this would still impact model performance; thus, future research will need to investigate this further.

There is a major limitation to this study due to a lack of standards used when collecting clinical features. Many of the variables in the UNITI tinnitus dataset were collected using subjective methods or in inconsistent ways between patients. Due to variability in recording protocols, variations in clinician interpretation, and differences in how patients report themselves, there is a high level of latent bias and noise that can't be corrected fully through preprocessing or feature selection. This lack of standardization makes it impossible to come close to achieving the theoretical performance ceiling; therefore there are limitations on the complexity of models developed. Compared with other benchmark datasets like DIA published by the UCI, which are consistently collected and cleaned for machine learning work, the use of real clinical datasets for machine learning applications will frequently require significantly more preprocessing than we will face when developing new models from benchmark datasets.

Another area, although it improves the development of models in some cases, is the use of SMOGN (synthetic minority oversampling technique) to conduct oversampling when generating new observations from the original dataset. When working on small datasets (e.g., $n \leq 50$), using synthetic augmentation could lead to a distorted distribution of the original data and could inadvertently create non-dependable-secrets through the creation of synthetic observations. When looking at the performance of models developed using the conservative parameters associated with SMOGN, decreased performance is frequently observed; thus, what would appear to be a gain from the use of oversampling may have come from the reinforcement of synthetic characteristics of the dataset rather than generalization to unknown data. As a result, results produced from oversampled datasets should be interpreted with caution.

Even though some feature selection procedures are capable of identifying predictive variables, there continues to be difficulties interpreting biological data, particularly as it pertains to genetic information. Just because a model meets the standards for prediction does not imply

that it has a causal relationships, and the diffused nature of the biological signal makes it next to impossible to relate the selected markers directly back to one specific biological pathway and without further validation from an expert in the field. By analyzing all feature selection methods used, the final THI score was not consistently predicted with the selected genetic data relative to the selected clinical and behavioral variables used in building the models. While this does not mean that there is no relationship between genetics and Tinnitus, it suggests that any relationship is probably very indirect and/or weakly captured in the current modeling method and sample size.

The generalizability issue is another large concern; because all models were built and validated on one dataset there may be differences in demographics and protocol in the external population that could impact the predictive accuracy of the model. A further validation study on independent datasets is warranted before any clinical applications can be realized.

Finally, both embedding model training and conducting large-scale experiments were very compute-intensive (i.e., considerable computing resources were used to train these models and execute large numbers of experiments), leading to limits on how many configurations/hyperparameters could be evaluated during the time frame of this project.

Experimental data imperfection simulations assisted in assessing the models under controlled settings, but may not represent the breadth of challenges encountered during clinical data collection. More extensive validation studies should be designed using naturally occurring imbalanced, noisy datasets.

In conclusion, although the models produced fairly low RMSE values (mostly around 10), in the context of THI scoring (1 to 100), this is still of clinically meaningful relevance. Due to the fact that a small difference in predicted value may change a patient’s severity classification, change the treatment recommended for the patient, or change how a clinician evaluates the patient, these models’ predicted performance can be statistically reasonable but not sufficient for direct clinical use without additional safeguards, uncertainty quantification and/or clinician supervision.

10 Conclusion

The goal of this project was to explore the use of predictive modeling based on ensemble methods and advanced techniques and pre-fitted tabular models for analyzing high-dimensional low sample space (HDLSS) tabular clinical data. To achieve this objective and ensure a reproducible experimental process, the authors developed a unified framework that covers all major stages of experimentation, including data simulation, feature extraction, model training, and performance evaluation.

The experiments presented in this report demonstrate that DeepFS and GRACES provide effective feature representations for both classification and regression tasks when combined with XGBoost and TabPFN models. In particular, their integration with TabPFN enabled robust learning under limited-sample and high-dimensional conditions, while behaviour XGBoost-based models offered strong baseline performance across different datasets.

Furthermore, controlled simulations of class imbalance and rare target values supported a real-world, deeper understanding of model behavior under uncertain and imperfect conditions data conditions. This systematic evaluation approach enabled reliable assessment of model stability, generalization capability, and practical applicability in real-world clinical scenarios.

11 Future Work

Our work can be further improved to better generalise to different use-cases and provide a better plug-and-play architecture. The first area is to make our work more scalable to provide reliable results for large and complex datasets which more diverse. Access to diverse and ample populations as well as heterogeneous data sources is associated with a more reliable and evaluation of the robust nature of our framework.

Secondly, advanced foundational models, along with large pre-trained tabular models, may lead to improved representation learning.

Future research could also revisit the application of large language models to tabular representation learning and develop methods to more effectively deal with cryptic and domain-specific column names which limited the utility of these methods in both studies.

The third area of potential future research relates to adaptive or dynamic feature selection methods that automatically adapt to evolving data distributions and missingness patterns. Adaptive/dynamic selection methods would be particularly beneficial to longitudinal clinical studies where patients experience changes in characteristics over time.

Additionally, more advanced simulation strategies may be designed to better represent real-world data collection processes. These simulations should consider elements such as temporal irregularities, measurement errors, and systematic biases; this is critical for the realistic evaluation of the methods used in this project.

12 Codebase and Documentation

The complete codebase developed for this project, along with the corresponding documentation for all utilities, preprocessing pipelines, and modeling frameworks, is available at the following GitHub repository: <https://github.com/BhaveshY/OptimisedKMDcode>.

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A Appendix

A.1 Original Assignment Description: “Prediction on High-Dimensional Clinical Data”

The core mission of our project, originally titled “Prediction on High-Dimensional Clinical Data,” was to build a specialized toolbox of embedding-based classifiers and regressors. Clinical research is notoriously difficult because it often involves **high-dimensional, multi-modal data** but only a handful of patient samples to work with. To tackle this, we mapped out a series of specific milestones.

First, we built a shared utility to ingest the clinical datasets and target variables. We then explored the existing literature to identify the best candidate methods for tabular data, while

simultaneously developing a simulator to evaluate model performance under increasing levels of missingness. Once we defined our evaluation criteria, we incorporated these methods into a reusable toolbox and benchmarked them through a custom evaluation suite.

Although the assignment was initially designed as a substantial task for a team of 3–4 experts in Python programming and deep learning, it provided a rigorous framework for testing the robustness and flexibility of our modeling pipeline.

A.2 Scope Adjustment

As the project moved forward, we had to face a hard truth: the original embedding-based plan looked great on paper but hit some serious roadblocks in practice. Finding specific models that could actually handle high-dimensional, low-sample size (HDLSS) data—while still meeting our strict technical criteria—was much tougher than we anticipated.

Rather than sticking to a failing plan, we decided to pivot and expand our scope to include algorithms that are actually built for these “wide but shallow” datasets. We integrated an embedding-based feature selector to dig into latent structures and added a similarity-based selector to capitalize on how patients relate to one another. By mixing these selectors with Transformer-based and tree-based models, we created a much more robust hybrid toolkit. This shift didn’t just help us bypass the data constraints; it ensured our final toolbox was actually useful for real-world clinical research.

A.3 Milestone Slides and TODOs

All milestone slides for the project, including the tasks and TODOs addressed after feedback, are available at the following GitHub repository: <https://github.com/Sajith-Santhosh/Milestone-Slides>.

A.4 TODOs post Milestone Presentation1

The main tasks covered in this milestone included:

TODO-1: Non-Pretrained Algorithm Selection

For the non-pretrained algorithms, we retained **GRACES** and **DeepFS**. Both methods are feature-name independent, designed for high-dimensional, low-sample size (HDLSS) datasets, and attentive to rare features. The older Deep Neural Pursuit (DNP) approach was excluded due to its outdated methodology, with newer approaches addressing its stability and feature correlation limitations.

TODO-2: Dataset and Pre-Trained Tabular Models

Regarding the DIA dataset, the cryptic feature names made typical large language model-based tabular algorithms difficult to apply. While no LLM-based algorithms were used in our setup, we retained pre-trained tabular models: **TabPFN** and **TABPTM**. TABPTM was kept for experiments because it effectively handles the DIA dataset by:

- Being feature-name independent, accommodating cryptic column names.
- Creating a shared vocabulary based on instance relationships (e.g., nearest neighbor distances) rather than feature names.
- Converting each instance into a fixed-dimensional meta-representation capturing local structure and neighborhood information.

These characteristics ensure TABPTM works effectively even when column semantics are inaccessible, aligning with the dataset requirements.

TODO-3: Simulator

The simulation phase, which generates variants of the original dataset to test robustness under increased missingness, has been included in the milestone slides.

A.5 TODOs Post Milestone 2

TODO-1: Uniti Dataset Collection

Following the second milestone presentation, the primary task was to collect the UNITI dataset after signing the necessary NDA. Once obtained, the dataset was reviewed to assess its structure, features, and potential issues, and any questions or doubts regarding the dataset were noted and reported. This step ensured that subsequent modeling efforts would be based on a clear understanding of the dataset and its limitations.

A.6 TODOs Post Milestone 3

TODO-1: Include All Utilities

All utilities developed during the project, including preprocessing utilities for methods that do not tolerate missingness, must be included in the toolbox to ensure reproducibility and full functionality.

TODO-2: Highlight Simulator

The simulator, which generates dataset variants under increased missingness, must be emphasized in the documentation and slides to showcase its role in testing model robustness.

TODO-3: Expand Evaluation Criteria

Evaluation must cover multiple criteria. While two methods may achieve similar prediction quality, differences in execution time, scalability, or resource requirements should also be captured to allow meaningful comparison.

TODO-4: Organize Results per Dataset

Results should be clearly presented for each dataset, including both the complete RCT dataset and the subset containing genetic data, to illustrate differences in model performance across data types.

TODO-5: RMSE Limitations and Confusion Matrix

Since RMSE values for the RCT dataset may be misleading, evaluation should also include a category-based assessment. Following Miro Schleicher's suggestion, count cases where the predictor places a patient in the correct category, one category away, two categories away, etc., effectively creating a confusion matrix for a more clinically meaningful assessment.